

The Second Workshop on Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI)

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ABSTRACT

With the advance of VR/AR technology, more and more VR/AR applications are emerging and have been popular among new users. Interacting with virtual reality and augmented reality technologies will require the development of alternative input modalities as well as coherent integration of multiple realistic sensations to increase the level of perceived realism. These developments will create a more immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. This workshop discusses the challenges and applications of designing a higher coherence between different input modalities and sensations to offer more engaging VR/AR experiences, which can create opportunities for the researchers from both UbiComp and VR/AR fields to jointly discuss and brainstorm alternative input modalities and sensations for VR/AR interactions.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

KEYWORDS

Virtual reality; Augmented reality; Multiple sensory modalities; Multi-sensations; Mixed reality; Realism; User friendliness.

ACM Reference Format:

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1 OBJECTIVE

With the advance of Virtual Reality and/or Augmented Reality (VR/AR) technology, more and more VR and AR apps are emerging and have been really very popular among new users [7]. To facilitate user interaction with the systems, multiple user input modalities are required. Furthermore, coherent integration of multiple realistic sensations can increase the level of realism [2], and therefore creates a truly immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. Therefore, many researchers are devoted their efforts to study how to achieve a higher level of coherence between different input modalities and sensations to provide more engaging VR/AR experiences.

Existing studies incorporate a wide range of input modalities (e.g., eye gaze [5], gesture [11], and voice [8]) to sense users' actions so that they can interact with virtual/augmented content as easily as they interact with the real world around them. Besides providing easy-to-use input modalities, many researchers explored ways to enhance realism in the VR/AR world to improve the level of immersiveness. Various sensations (e.g., visual perception[3], auditory cues [4], haptic feelings [10], and olfactory sensations [6]) were incorporated into different VR/AR applications. Leveraging the advances of a combination of VR/AR software and hardware, VR/AR user experience has greatly improved recently, which attracts an increasing number of interdisciplinary researchers, designers, and engineers to apply VR/AR technology into various application domains. However, due to insufficient support of input modalities or sensations, current VR/AR interactions are still not able to provide users with realistic and immersive feelings. To continue improving VR/AR user experience, this workshop explores to bring in more talented UbiComp researchers on studying how to design better input modalities and sensations.

In order to gather research momentum from the UbiComp community, we call for papers discussing how to design or incorporate different input modalities or sensations for enabling a more immersive VR/AR experience. To reduce long-haul travel following the pandemic, MIMSVAI 2022 will be a virtual workshop held on September 11th for on-site as well as online attendees, following a half-day format. Those expecting to attend the UbiComp/ISWC 2022 conference may also register during this workshop. We arrange sessions for each accepted paper to be presented in the workshop.

At the end of each session, we will assign session chairs to mediate the discussion to distill motivated ideas and findings to the participants. Besides the regular sessions, we will invite one expert for VR/AR interactions to give talks in the morning. We hope the keynote presentation can bring in more interesting perspectives regarding VR/AR-related researches. Finally, we will deliver the closing remarks and give the best paper award to the best paper presented at the MIMSVAI 2022 workshop to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

2 RELEVANT AND TIMELY

Ubiquitous Computing (UbiComp) has always been a field that focuses on bringing computing to people's physical world. Mark Weiser claimed that UbiComp and virtual reality are fundamentally opposed. Ubiquitous computing entails placing wearable computing devices in the real world, whereas virtual reality places people within a computer-generated world [9]. On the contrary, Virtual Reality (VR) simulates the physical world and modifies human sensory input to put people in the virtual computing world. In between these two extremes, augmented reality (AR) overlays a virtual world over the real world as a complement to our sensation of the ambient environment. AR plays an intermediate role to enrich, rather than replace the physical world with mobile or wearable devices.

As the technology advances in the UbiComp or VR/AR field, new possibilities of applying the combinations of these technologies in different domains emerges, which blurs the line between UbiComp and VR/AR technologies. VR/AR is melding with UbiComp field. UbiComp technology interacts with users through multiple sensory and feedback technologies in the physical world through mobile and wearable devices. The same sets of sensory and feedback technologies can be used to design multiple input modalities and sensations for users to immersively interact with both physical and virtual worlds through VR/AR devices. Moreover, both UbiComp and VR/AR fields require Human-Computer Interaction (HCI) methodologies to incorporate the designed solutions into different application domains to increase their impact. It is important to consider that both systems require human-computer interaction (HCI) technologies, which means that developments in one field could be applied in the other. Therefore, we expect this workshop can create opportunities for the researchers from both UbiComp and VR/AR fields to jointly discuss and brainstorm new directories of designing new input modalities and sensations for VR/AR interactions.

3 TOPICS

The workshop fosters discussions among UbiComp and VR/AR fields to cover (1) improving input modalities, (2) creating sensations of greater realism, (3) developing alternative devices and infrastructures, (4) identifying novel application scenarios, and (5) expanding design process, platform/tools, or libraries for enabling immersive VR/AR interactions. We expect contributions (but not limited to) on the following topics:

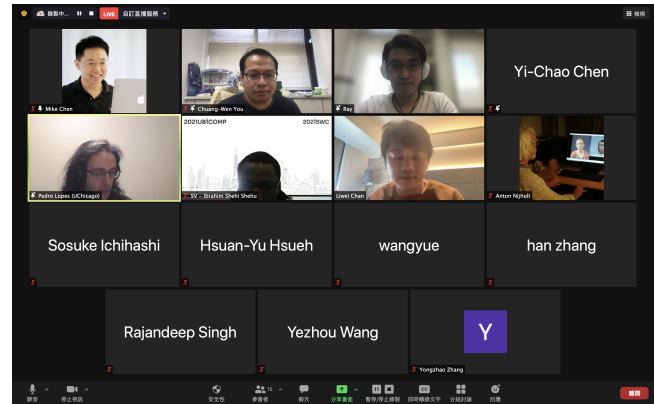


Figure 1: Photos from MIMSVAI 2021 (Virtual Event).

- **Improving input modalities.** This will require the sensing of user actions/intentions during interactions with the virtual world. Existing input modes will have to be improved, abandoned in favor of other input modes, or combined with other input modes.
- **Creating sensations of greater realism.** This will involve giving the user a realistic impression of the sensations one would expect to experience when dealing with specific virtual phenomena. It is likely that this will involve expanding existing methods with entirely new methods.
- **Developing alternative devices and infrastructures.** Researchers will have to envision entirely new approaches to VR/AR devices and infrastructures that are not limited to the computational limits of first-generation devices.
- **Identifying novel application scenarios.** VR/AR is already used in gaming, healthcare, and education; however, there is little doubt that 5 to 10 years will pass before these immersive technologies approach maturity. In the future, this technology will be applied in ways that are beyond our imagination at this point. It is time for researchers to assume a visionary role in expanding the range of applications to which these new technologies can be applied.
- **Expanding design process, platform/tools, or libraries.** This will require anticipating user needs and the ways that those needs can be satisfied. Designing new VR/AR applications will require profound changes in existing platforms/tools (e.g., Unity) and libraries (e.g., Android ARCore or Apple ARKit) as well as the development of entirely new modalities.

4 WORKSHOP STRUCTURE

Following the previous successful MIMSVAI 2021 workshop (shown in Figure 1), MIMSVAI workshop will be a virtual workshop held on September 11th for on-site as well as online attendees, following a half-day format. Those expecting to attend the UbiComp/ISWC 2022 conference may also register during this workshop. We plan different activities during the workshop to make it more interactive, including the introduction to the workshop, one keynote talk, multiple paper presentations, and one best paper award (see Table 1). Particularly, we will invite one expert for VR/AR interactions to give

a talk so the workshop can bring in more interesting perspectives regarding VR/AR-related researches in the UbiComp community. Finally, we will deliver the closing remarks and give the best paper award to the best paper presented at the MIMSVAI 2022 workshop to acknowledge and encourage excellence in research.

Table 1: Workshop Schedule

Workshop Schedule	
9:00-9:10 am	Opening
9:10am-10:20am	Keynote Talk I: 50 + 20min
10:20am-10:30am	Coffee Break
10:30am-11:10am	Paper Session I: $2 \times (15 + 5)\text{min}$
11:10am-11:20am	Coffee Break
11:20am-12:00pm	Paper Session II: $2 \times (15 + 5)\text{min}$
12:00pm-12:10pm	Best Paper Award

Papers submitted for presentation during the workshop must not exceed 5 pages in length (4-pages plus 1 page for references) in accordance with the ACM sigconf template [1]. The accepted papers will be published in the UbiComp/ISWC Adjunct Proceedings, which will be included in the ACM Digital Library. The Best Paper Award will be given to the best paper presented at the MIMSVAI 2022 workshop, to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

The call for papers will be sent to all relevant mailing lists of several research communities, including those of UbiComp research, HCI, VR/AR applications, haptic and tactile interfaces, and multi-sensory perception communities. The submissions will be peer-reviewed by no fewer than two reviewers drawn from PC members and organization staff. Our acceptance criteria will be a mixture of novelty, research quality, and relevance. It is expected that four papers will be accepted for presentation. Given the theme of this workshop and the rapid growth of the VR/AR community, a large number of submissions are expected which enables us to organize a high-quality workshop.

5 EXPECTED OUTCOMES

The workshop is open to individuals who submit papers as well researchers with an interest in VR/AR technology. We expect to attract no fewer than 15 participants. It is hoped that this workshop will enhance our understanding of VR/AR interactions in terms of state-of-the-art methods and changes anticipated for the future.

ACKNOWLEDGMENTS

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6 APPENDIX: CALL FOR PAPERS

With the advance of VR/AR technology, more and more VR/AR applications are emerging and have been popular among new users. Interacting with virtual reality and augmented reality technologies will require the development of alternative input modalities as well as coherent integration of multiple realistic sensations to increase the level of perceived realism. These developments will create a more immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. This workshop discusses the challenges and applications of designing a higher coherence between different input modalities and sensations to offer more engaging VR/AR experiences, which can create opportunities for the researchers from both UbiComp and VR/AR fields to jointly discuss and brainstorm alternative input modalities and sensations for VR/AR interactions.

Papers may include, but not be limited to, topics such as:

- 2D/3D and volumetric display and projection technology
- Immersive analytics and visualization
- Modeling and simulation
- Multimodal capturing and reconstruction
- Scene description and management issues
- Storytelling
- Tracking and sensing
- Audio interfaces, sound rendering, spatialized audio, auditory perception and psychoacoustics
- Embodied agents, virtual humans and (self-)avatars

- Haptic and tactile interfaces, wearable haptics, passive haptics, pseudo haptics, other touch-based UI
- Mediated and diminished reality
- Multimodal input and output
- Multisensory rendering, registration, and synchronization
- Perception and cognition
- Presence, body ownership, and agency
- Teleoperation and telepresence
- 3D user interaction
- 3D UI metaphors
- Collaborative interactions
- Ethical issues
- Human factors and ergonomics
- Input devices
- Locomotion and navigation
- Multimodal/cross-modal interaction and perception
- Non-fatiguing 3D UIs
- Non-visual interfaces (such as olfactory)
- Touch, tangible and gesture interfaces
- Usage research, evaluation methods and empirical studies
- Domain-specific VR/AR applications.

- Design process, platform/tools, or libraries used to incorporate input modalities or sensations for VR/AR interactions or studying their use

Important Dates

Paper submission: July 29, 2022

Paper notification: August 21, 2022

Camera-ready: TBA

Workshop: September 11, 2022

Best Paper Award

The Best Paper Award is given to the best paper presented at the MIMSVAI 2022 workshop, to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

Workshop Organizer

Chuang-Wen You, National Tsing Hua University

Yi-Chao Chen, Shanghai Jiao Tong University

Hsin-Ruey Tsai, National Chengchi University

Chu-Yin Chen, Paris 8 University